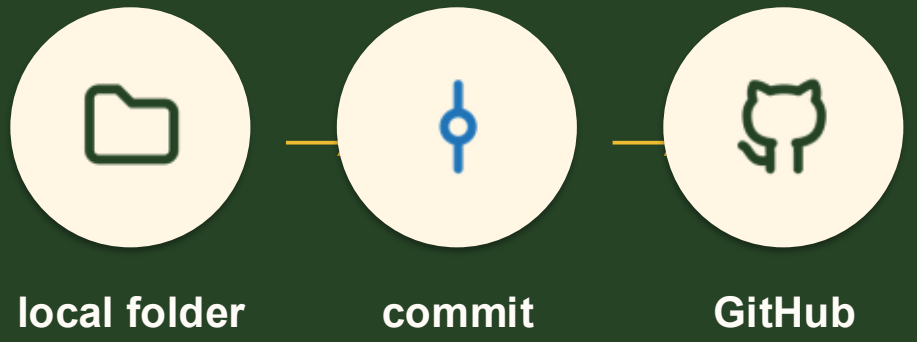
Git & GitHub

Module 0: Foundations



By the End of Today, You'll Be Able To…





01

Start a local git repository



02

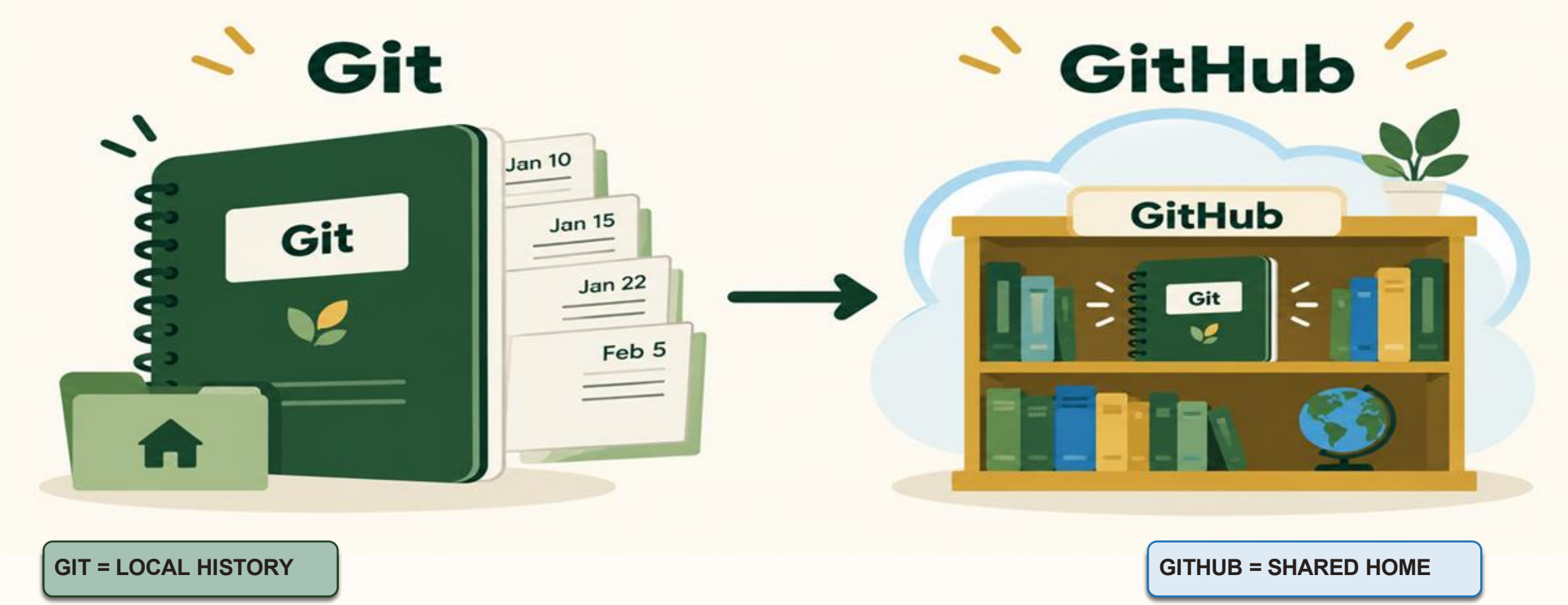
Make a well-messaged commit



03

Push your first commit to GitHub

Git Is a Notebook; GitHub Is the Shared Shelf



GIT = LOCAL HISTORY

GITHUB = SHARED HOME

Key Terms



Repository

A project folder git is tracking

Commit

A saved snapshot at one point in time

Remote

A hosted copy, like GitHub

TERMS NAMED

Code Along: Start Tracking a Project



A repository is just a folder git is watching — let's actually make one.

Start Tracking a Project



bash

```
$ mkdir my-first-repo
$ cd my-first-repo
$ git init
```

Run this together. Then use **ls -a** to see the hidden .git folder.

A Repo Needs a README





Why it matters

README.md explains what this project is

It's the first thing anyone else reads
Write it before you forget why

README.md

```
# My First Repo

This project is a
place to practice
Git and GitHub.
```

README WRITTEN

Code Along: Stage and Check Status



Writing the file is one thing — git doesn't know it exists until you tell it to.

Stage and Check Status



bash

```
$ touch README.md
$ git add README.md
$ git status
```

git status is safe to run any time. Build the habit before and after each command.

Make Your First Commit



bash

```
$ git commit -m "Add initial README"
```

A specific commit message tells future-you what changed.

Not this: “update”

Do this: “Add initial README”

See Exactly What Changed



bash

```
$ echo "More notes coming soon." >> README.md
$ git diff
```

git diff shows the actual line-level changes before the next commit.

YOUR LOCAL REPO

Connecting to GitHub



Everything so far has lived only on your machine — nothing's been shared with anyone yet.

Making It Public: GitHub





Create

a new, empty repository on github.com



Copy

the remote URL it gives you



Skip

“Add a README” — yours exists

REMOTE CREATED

Code Along: Link and Push



An empty GitHub repo and your local commits still don’t know about each other yet — this connects them.

Link and Push



bash

```
$ git remote add origin <your-repo-url>
$ git push -u origin main
```

This is the moment your local work becomes visible on GitHub.

See Your Own History



bash

```
$ git log --oneline
```

The payoff: a readable history you built yourself.

A good log is a project timeline, not a mystery.

Common First-Day Errors





"not a git repository"

You're in the wrong folder



"nothing to commit"

You probably forgot git add



Password prompt

GitHub wants a token

CHECK FOR UNDERSTANDING

Which command tells you what's actually going on?



You just ran a command and nothing seemed to happen.



A

git init



B

git status



C

git push

ANSWER KEY

When Stuck, Check Status

git status

It’s safe to run and tells you the current state: staged files, unstaged changes, and what branch you’re on.

ADD TO THE BOARD

Put the Git Steps in Order



A

git push

B

git commit

C

git add

D

git init

Post them in the order you'd actually run them.

CHECK FOR UNDERSTANDING

Rewrite this commit message: "update"



A

"stuff"

B

"final update 2"

C

"Fix typo in README installation steps"

ANSWER KEY

Rewrite a Bad Commit Message



"Fix typo in README installation steps"

A good message names the specific what. Someone reading the log later should understand the change without opening the diff.

Key Terms — Recap



Repository	A project folder git is tracking the history of
Commit	A saved snapshot of your project at one point in time
Remote	A copy of your repository hosted somewhere else, like GitHub

What You Have Now



A real local repo
with real commits

That repo
pushed to GitHub

The habit
of checking git status often

Before Next Class



Keep

your practice repo around

Next class

Same sequence, on your own

Lab

Git & Terminal Practice

KEEP LEARNING

Where to Go From Here



Git Basics

Pro Git book, Ch. 2

DOCS · 15–20 min

Free canonical git reference.

GitHub Docs: Hello World

PRACTICE · 15 min

GitHub’s short guided first-repo walkthrough.